

Chapter 1. Vectors

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Preliminaries / notation:

$\mathbb{N}$ - natural numbers (e.g. "0", 1, 2, 3, ...)

$\mathbb{Z}$ - integers (e.g. ..., -3, -2, -1, 0, 1, 2, 3, ...)

$\mathbb{Q}$ - rational numbers, i.e. $\frac{p}{q}$ for $p \in \mathbb{Z}$ and $q \in \mathbb{Z} \setminus \{0\}$
(e.g. -2, $\frac{3}{4}$, 0.001, $\frac{7}{9}$, ...)

$\mathbb{R}$ - real numbers (e.g. 1, π , $\sqrt{2}$, $\frac{7}{8}$, e, ...)

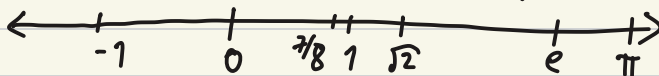
$\mathbb{R}$ includes rational numbers,
and irrational numbers.

Every real number has a decimal expansion

Rational numbers have terminating or repeating expansions

e.g. $\frac{7}{8} = 0.875$ $\frac{7}{9} = 0.\overline{7}$
 $\pi = 3.14159265\dots$

Use the real line to represent real numbers
(larger numbers go further to the right)



Other set notation: $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$ union
 $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$ intersection
 $A \setminus B = \{x \mid x \in A \text{ but } x \notin B\}$ difference

e.g. say $A = \{1, 3, 6, 11\}$ and $B = \{3, 5, 9, 10\}$.
 $A \cup B = \{1, 3, 5, 6, 9, 10, 11\}$
 $A \cap B = \{3\}$ $A \setminus B = \{1, 6, 11\}$ $B \setminus A = \{5, 9, 10\}$

1.1 Vectors in two and three dimensions

def: A vector in $\mathbb{R}^2$ is an ordered pair of real numbers.

$$\vec{a} = (a_1, a_2) \quad \mathbb{R}^2 = \{(x, y) \mid x, y \in \mathbb{R}\}$$

def: A vector in $\mathbb{R}^3$ is an ordered triple of real numbers.

$$\vec{a} = (a_1, a_2, a_3) \quad \mathbb{R}^3 = \{(x, y, z) \mid x, y, z \in \mathbb{R}\}$$

we can also go to n -dimensions, but we will rarely go beyond 3 in this class

... a vector in $\mathbb{R}^n$ is an ordered n -tuple of real numbers

$$\vec{a} = (a_1, a_2, \dots, a_n) \quad \mathbb{R}^n = \{(a_1, \dots, a_n) \mid a_1, \dots, a_n \in \mathbb{R}\}.$$

Observation: Two vectors are equal if their components are equal

$$\text{e.g. } (2, 3) = (\frac{8}{4}, \sqrt{27})$$

Operations with vectors (addition and scalar multiplication):

We only add vectors of the same dimension componentwise

$$\begin{array}{l} \text{2-D} \\ \vec{a} + \vec{b} = (a_1, a_2) + (b_1, b_2) = (a_1 + b_1, a_2 + b_2) \\ \vec{a} = (a_1, a_2) \quad \vec{b} = (b_1, b_2) \end{array}$$

$$\begin{array}{l} \text{3-D} \\ \vec{c} + \vec{d} = (c_1 + d_1, c_2 + d_2, c_3 + d_3) \\ \vec{c} = (c_1, c_2, c_3) \quad \vec{d} = (d_1, d_2, d_3) \end{array}$$

We can multiply a vector $\vec{a}$ by a scalar $k \in \mathbb{R}$ by multiplying each component of $\vec{a}$ by the scalar k .

$$\begin{aligned}\vec{a} &= (a_1, a_2) & k \vec{a} &= (or\ k \cdot \vec{a}) = (k a_1, k a_2) \\ \vec{b} &= (b_1, b_2, b_3) & k \vec{b} &= k \cdot \vec{b} = (k b_1, k b_2, k b_3)\end{aligned}$$

Properties of vector addition (in $\mathbb{R}^n$)

commutativity : $\vec{a} + \vec{b} = \vec{b} + \vec{a}$ for all $\vec{a}, \vec{b} \in \mathbb{R}^n$

associativity : $\vec{a} + (\vec{b} + \vec{c}) = (\vec{a} + \vec{b}) + \vec{c}$ for all $\vec{a}, \vec{b}, \vec{c} \in \mathbb{R}^n$

zero vector : there is a vector $\vec{0}$ such that $\vec{a} + \vec{0} = \vec{a}$ for all $\vec{a} \in \mathbb{R}^n$.

Properties of scalar multiplication

distributivity : $(k+l) \vec{a} = k \vec{a} + l \vec{a}$ for all $k, l \in \mathbb{R}$ and $\vec{a} \in \mathbb{R}^n$

distributivity : $k(\vec{a} + \vec{b}) = k \vec{a} + k \vec{b}$ for all $k \in \mathbb{R}$ and $\vec{a}, \vec{b} \in \mathbb{R}^n$

associativity : $k(l \vec{a}) = (kl) \vec{a}$ for all $k, l \in \mathbb{R}$ and $\vec{a} \in \mathbb{R}^n$

identity : $1 \vec{a} = \vec{a}$ for all $\vec{a} \in \mathbb{R}^n$

Geometric notion

We use the real line $\longleftrightarrow$ We use real numbers
to represent the real numbers $\longleftrightarrow$ to give coordinates on the line

points in a line $\longleftrightarrow$ real numbers (vectors in $\mathbb{R}^1$)

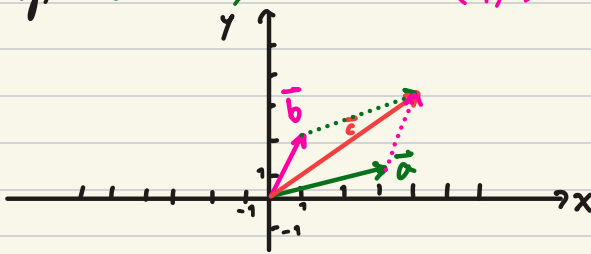
points in a plane $\longleftrightarrow$ vectors in $\mathbb{R}^2$

points in 3D space $\longleftrightarrow$ vectors in $\mathbb{R}^3$

e.g. $\vec{a} = (3, 1)$

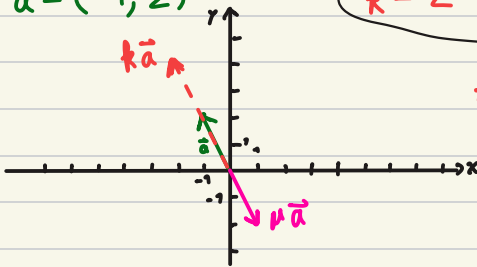
$\vec{b} = (1, 2)$

$\vec{c} = \vec{a} + \vec{b} = (3+1, 1+2)$
 $= (4, 3)$



vector addition corresponds to the diagonal of the parallelogram

e.g. $\vec{a} = (-1, 2)$

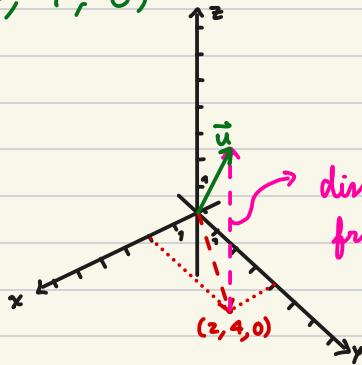


$k = 2$ $\mu = -1$ $\rightarrow$ scalars

$k\vec{a} = (2 \cdot (-1), 2 \cdot (2)) = (-2, 4)$

$\mu\vec{a} = (-1 \cdot (-1), -1 \cdot (2)) = (1, -2)$

e.g. $\vec{u} = (2, 4, 6)$



displacement vector $\vec{P_1 P_2} = (0, 0, 6)$
 from $P_1(2, 4, 0)$ to $P_2(2, 4, 6)$

$\rightarrow$ points in space

To emphasize the difference between points in space and vectors, we sometimes denote points by $P(x, y, \dots)$. We reserve the notation $\vec{a}, \vec{b}, \dots$ for vectors.

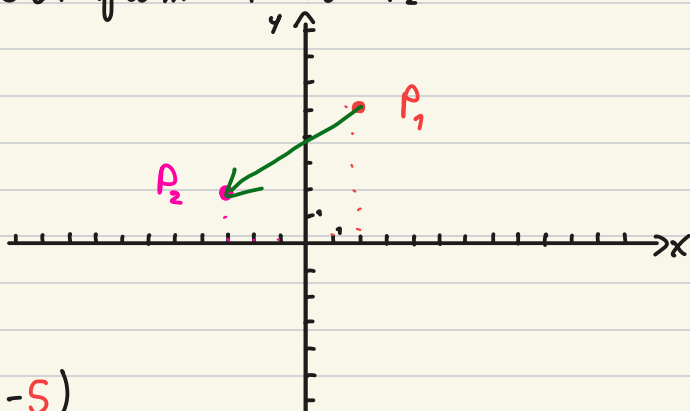
e.g. Displacement vector from P_1 to P_2

$$P_1 = (2, 5)$$

$$P_2 = (-3, 2)$$

$$\overrightarrow{P_1 P_2} = (-5, -3)$$

$$\vec{P}_2 - \vec{P}_1 = (-3 - 2, 2 - 5)$$



e.g. Particle travelling in a straight-line through space
(Figure 1.13 in book)

initial position vector $\vec{a} = \vec{x}_0 = (2, 5, 4)$

velocity vector $\vec{v} = (4, 2, -1)$

position after a time $t = 2$

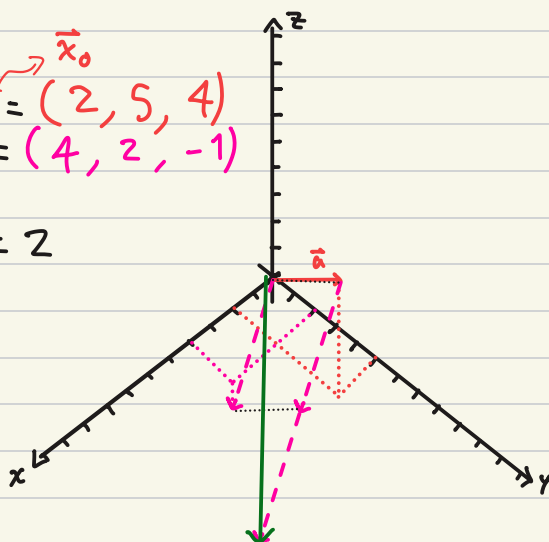
$$\vec{x} = \vec{a} + t \vec{v}$$

$$\vec{x} = \vec{x}_0 + t \vec{v}$$

$$= (2, 5, 4) + 2(4, 2, -1)$$

$$= (2, 5, 4) + (8, 4, -2)$$

$$= (10, 9, 2)$$



1.2. More about vector

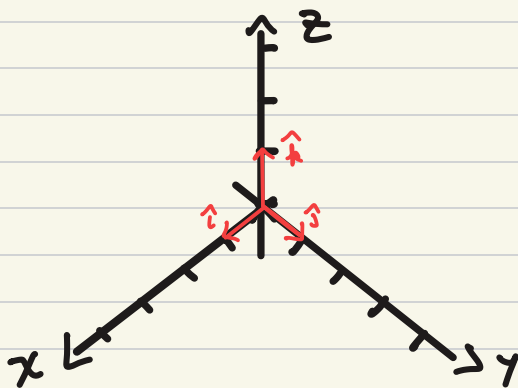
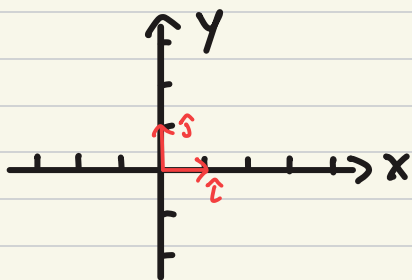
In $\mathbb{R}^2$ a vector $\vec{a} = (a_1, a_2)$ can be written in terms of the special unit vector $\hat{i} = (1, 0)$ and $\hat{j} = (0, 1)$

$$\vec{a} = (a_1, a_2) = a_1 \hat{i} + a_2 \hat{j}$$

def: $\hat{i}$ and $\hat{j}$ are called the standard basis vector in $\mathbb{R}^2$

Moving to $\mathbb{R}^3$ we increase the dimension by one, so we need one more standard basis vector

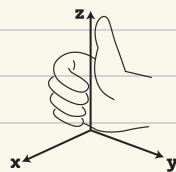
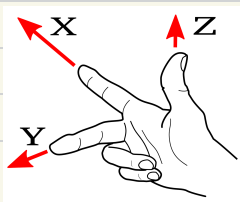
standard basis vector $\hat{i} = (1, 0, 0)$, $\hat{j} = (0, 1, 0)$, $\hat{k} = (0, 0, 1)$



About orientation of graph axes

2-D: as we are used to (y-axis adjacent counter-clockwise $\curvearrowright$ to x-axis)

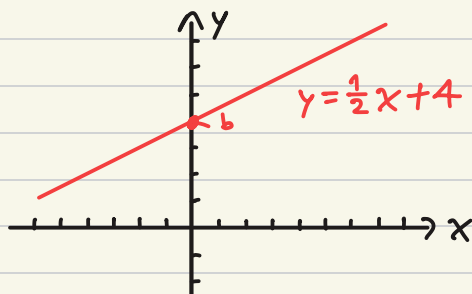
3-D: right hand rule



Graphing lines

- In 2-D: a single linear equation

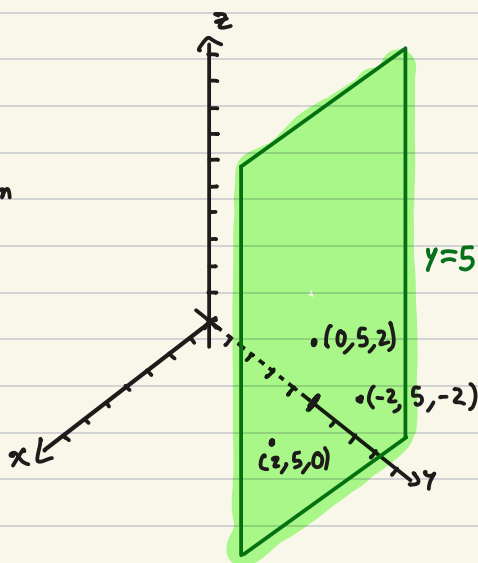
$$\text{or } \begin{aligned} Ax + By &= C \\ y &= mx + b \end{aligned}$$



- In 3-D:

a single linear equation in 3-D defines a plane

$$Ax + By + Cz = D$$



- Parametric equations of a curve

$$\text{in 2-D } \begin{cases} x = f(t) \\ y = g(t) \end{cases}$$

(parameter t)

from above $y = \frac{1}{2}x + 4$

$$x = 2t$$

$$y = t + 4$$

verify

$$\frac{1}{2}x + 4 = \frac{1}{2}(2t) + 4 = t + 4$$

$$\text{eg. 2: } \begin{cases} x = t \\ y = \frac{1}{2}t + 4 \end{cases}$$

from parametric to slope-intercept:

1- solve for t in x -equation

$$\rightarrow t = \frac{x}{2}$$

2- plug t into y -equation

$$\rightarrow y = t + 4 = \frac{x}{2} + 4$$

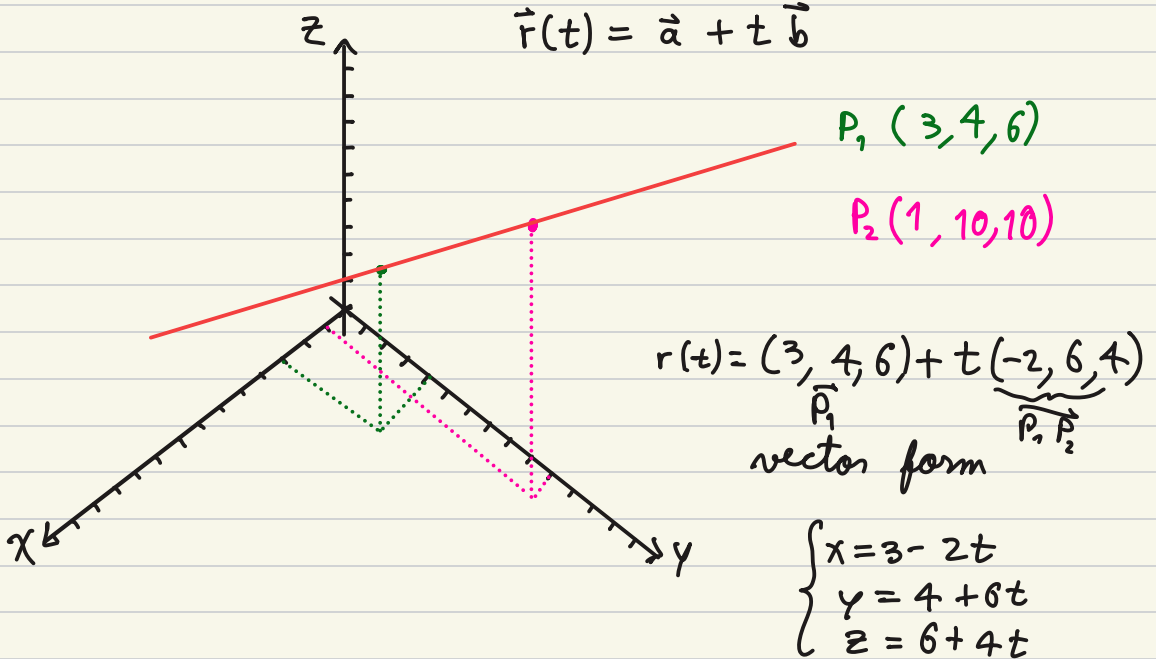
- Parametric equations of a curve in 3-D $\begin{cases} x = f(t) \\ y = g(t) \\ z = h(t) \end{cases}$

Vector equation of a line (in any dimension)

line is determined by two points P_1 P_2
 $\vec{x} = \vec{P}_1 + t \vec{P_1 P_2}$

or a starting point $\vec{b}$ and a direction vector $\vec{a}$

$$\vec{r}(t) = \vec{a} + t \vec{b}$$



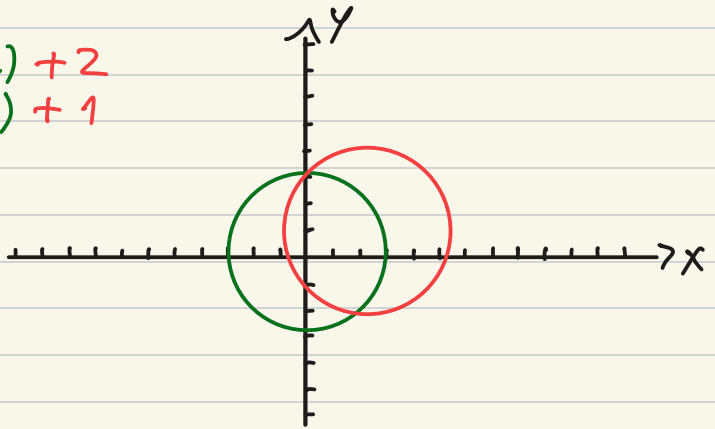
Symmetric form of the equation of a line

$$\frac{x-x_0}{a} = \frac{y-y_0}{b} = \frac{z-z_0}{c}$$

$t = \frac{x-3}{-2}$ $t = \frac{y-4}{6}$ $t = \frac{z-6}{4}$

$\frac{3-x}{2} = \frac{y-4}{6} = \frac{z-6}{4}$

e.g.
$$\begin{cases} x = 3 \cos(t) + 2 \\ y = 3 \sin(t) + 1 \end{cases}$$



e.g.
$$\begin{cases} x = 4 \cos t \\ y = 4 \sin t \\ z = t/\pi \end{cases}$$

plot over the domain
 $t \in [0, 4\pi]$

